Spectral matching on a learned background

Detection, identification and abstention for sub-pixel targets in the Allotrope residual

Allotrope research note · research/math/spectral-matching

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Abstract

This note sets out the mathematics of naming the material of a flagged anomaly when the background is predicted by a masked foundation model (Indradhanu Heavy) rather than estimated from a ring of neighbours. It derives the residual model, shows that residual Mahalanobis (RM) is a generalised likelihood ratio test for an unrestricted signal and that family matching is its restriction to library subspaces. It then treats known-target detection and fill estimation on the residual, families as subspaces, the leakage of the target into its own predicted background, calibrated abstention, the size of each error source, a learned namer as a Bayes classifier under a generator (with the namer built on 31 October 2026 set against it), and the sample sizes a material benchmark needs. Numbers come from this repository and are tagged: [M] measured in a project experiment (path given), [C] computed for this note by `compute_numbers.py` from the shipped library and constants, [A] assumed.

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Sources read and relied on. Where a statement rests on one of these the path is given in the text. I checked each cited constant against the code (window sizes, sample sizes, $d = 30$, the p^* rule, the code widths D) and say so where it matters.

- Plans: `docs/proposals/spectral-matching-fresh-start.md` (parked plan), `docs/proposals/spectral-matching-foundation-model.md` (earlier proposal), `docs/lld/spectral-family-match.md`, `spectal_match_sample/WALKTHROUGH.md`.
- Code: `app/utils/anomaly_detection/residual_mahalanobis.py`; `app/spectral_match/` (`family_match.py`, `sam.py`, `resample.py`); `backend/allotrope/action_types/_spectral_library_match_run.py`; `app/foundation_models/components/` (`pixel_normalization.py`, `spectral_reduced.py`); `app/evaluation/implant.py`, `app/evaluation/significance.py`.
- Experiments under `research/experiments/`: `indradhanu-finetune/FT06-uncertainty`, `FT07-gated` (records FT08), `FTH1-heavy-joint`, `FTH3-heavy-joint-more2`; `decoder-capacity`, `bench-resolution`, `atm-correction` (each its `README.md`).

1 Setting and notation

1.1 Sensors, bands and normalisation

A pixel spectrum is a vector $\mathbf{x} \in \mathbb{R}^N$ sampled at band centres $\lambda_1, \dots, \lambda_N$ with full widths at half maximum $w_1, \dots, w_N$. The native band sets differ: PRISMA 234 bands in the shards (239 in the vendable before filtering), EnMAP 224, AVIRIS-NG 372 good bands, SpectralEarth 202 (`docs/03-data-pipeline.md`). The product's common grid is 165 bands at 10 nm over 460–2450 nm with the water and CO_2 windows removed (`app/models/dataset/vendables.py`, `build_common_wavelength_grid`). Indradhanu reads native bands through its wavelength-aware stem, so the analysis below is written for a generic N; numerical examples use the 165-band grid.

Definition 1.1 (Normalisation by wavelength)**.** Let $\mu(\lambda)$ and $\sigma(\lambda)$ be the mean and standard deviation curves of surface reflectance, measured once on SpectralEarth's training split and stored in the checkpoint (`WavelengthStats`, `pixel_normalization.py`). For a band list, $\boldsymbol{\mu} = (\mu(\lambda_n))_n$, $\mathbf{D} = \operatorname{diag}(\sigma(\lambda_n))$, with linear interpolation between measured wavelengths and end values held outside them. The normalised spectrum is

$$\mathbf{z}(\mathbf{x}) = \mathbf{D}^{-1}(\mathbf{x} - \boldsymbol{\mu}).$$

The map $\mathbf{z}$ is affine and the same for every scene, every pixel and every library spectrum. Two consequences are used throughout.

Proposition 1.2 (Mixing survives normalisation)**.** *If* $\mathbf{x} = a\mathbf{t} + (1-a)\mathbf{b} + \mathbf{n}$ *then* $\mathbf{z}(\mathbf{x}) = a\,\mathbf{z}(\mathbf{t}) + (1-a)\,\mathbf{z}(\mathbf{b}) + \mathbf{D}^{-1}\mathbf{n}$*, and every difference transforms linearly:* $\mathbf{z}(\mathbf{t}) - \mathbf{z}(\mathbf{b}) = \mathbf{D}^{-1}(\mathbf{t} - \mathbf{b})$.

Proof. $\mathbf{D}^{-1}(a\mathbf{t} + (1-a)\mathbf{b} + \mathbf{n} - \boldsymbol{\mu}) = a\mathbf{D}^{-1}(\mathbf{t} - \boldsymbol{\mu}) + (1-a)\mathbf{D}^{-1}(\mathbf{b} - \boldsymbol{\mu}) + \mathbf{D}^{-1}\mathbf{n}$ because $a + (1-a) = 1$. The mean cancels in a difference. □

Per-scene, per-band min-max scaling is also affine within one scene, but its coefficients depend on that scene's extremes, so a library spectrum has no defined image under it; the earlier proposal records this as the reason HyperSL cannot be used for library matching. Under Definition 1.1 the library and the image are mapped by the same operator, which is what makes residual-domain matching possible at all. For scale: on the 165-band grid the reflectance standard deviation runs from 0.035 (blue) to 0.100, median 0.077, and the mean curve has median 0.155 (`app/constants/hyperspectral.json`)[C]. One normalised unit is therefore about 0.08 reflectance. This file is the 165-grid statistic, a stand-in for SpectralEarth's `stats.json`, which is not in the repo.

1.2 The library through the sensor

A library entry is a reflectance curve $\rho(\lambda)$ on the 2151-channel ASD axis (350–2500 nm, `data/family_library/`). The sensor's band n integrates it against its spectral response s_n:

$$t_n = \frac{\int s_n(\lambda)\rho(\lambda)\,d\lambda}{\int s_n(\lambda)\,d\lambda}, \qquad s_n(\lambda) = \exp\Big(-\frac{(\lambda-\lambda_n)^2}{2\varsigma_n^2}\Big), \quad \varsigma_n = \frac{w_n}{2\sqrt{2\ln 2}},$$

truncated at $3\varsigma_n$ and set to NaN if fewer than three library samples fall in the window (`app/spectral_match/resample.py`; derivation in `WALKTHROUGH.md` chapter 2). In matrix form $\mathbf{t} = \mathcal{S}\rho$ with $\mathcal{S}$ row-stochastic. Because $\mathcal{S}$ is linear and row-stochastic, it commutes with mixing: resampling a mixture equals mixing the resampled spectra. The shipped family library has 411 spectra in 17 families, 176 man-made in 10 families (counted from the directory; it matches the LLD); 352 of the 411 are finite on all 165 grid bands at 10 nm FWHM[C].

1.3 The mixing model and the point-spread function

Model 1.3 (Linear sub-pixel mixing)**.** At pixel p,

$$\mathbf{x}_p = a_p\,\mathbf{t} + (1 - a_p)\,\mathbf{b}_p + \mathbf{n}_p, \qquad a_p \in [0, 1],$$

with $\mathbf{t}$ the target spectrum, $\mathbf{b}_p$ the local background (what p would show without the target), a_p the fill and $\mathbf{n}_p$ sensor noise, $\mathrm{Cov}(\mathbf{n}_p) = \Sigma_n$.

The fill is not the geometric area fraction. With a target occupying the ground region Ω and a system point-spread function h (pixel aggregation convolved with optics and motion),

$$a_p = \int_\Omega h(\mathbf{c}_p - \mathbf{u})\,d\mathbf{u},$$

so one object produces a field of fills $\{a_q\}$ over a neighbourhood. The implant model (`app/evaluation/implant.py`, `psf_weights`) uses a Gaussian of standard deviation $\sigma_{\mathrm{PSF}} = 1.4$ px, normalised so that the footprint's mean weight is one.

Worked example 1.4 (How far a target spreads)**.** With $\sigma_{\mathrm{PSF}} = 1.4$ px a target filling exactly one pixel delivers only 7.5% of its energy to that pixel; the rest spreads over roughly a 5×5 neighbourhood[C]. Under `psf_weights`' normalisation the mean weight on the one-pixel halo is 0.78 for a 1×1 object, 0.64 for 2×2 and 0.49 for 4×4[C]. The comment in `implant.py` records that real PRISMA bench objects have a halo-to-object contrast of 0.74[M], so $\sigma = 1.4$ reproduces the halo of small real objects. Read literally as a sensor PSF (FWHM $\approx$ 3.3 px, about 100 m), that is pessimistic for PRISMA; part of the measured halo is probably label misregistration (the bench ground truth may be offset by about one pixel). I keep $\sigma = 1.4$ as an effective spread, not a claim about the optics.

2 The learned background

2.1 The hidden-pixel prediction and the residual

Indradhanu is a masked autoencoder. In the product, each pixel is predicted in one of two complementary checkerboard passes over 4×4-pixel tokens, from a context $C(p)$ that excludes p's own token (`spectral_stem_inferencer.py`, `_checkerboard_keep_mask`; `decoder-capacity/visible_vs_hidden.py` calls this "the product two-pass"). Write

$$\hat{\mathbf{b}}_p = f_\theta\big(\mathbf{x}_{C(p)}\big).$$

Definition 2.1 (Residual and background-prediction error)**.** $\mathbf{r}_p = \mathbf{x}_p - \hat{\mathbf{b}}_p$ is the residual and $\mathbf{e}_p = \mathbf{b}_p - \hat{\mathbf{b}}_p$ the background-prediction error, with $\Sigma_e = \mathrm{Cov}(\mathbf{e})$ over background pixels.

Substituting Model 1.3,

$$\mathbf{r}_p = a_p(\mathbf{t} - \mathbf{b}_p) + \mathbf{e}_p + \mathbf{n}_p. \tag{1}$$

An equivalent form that avoids the unknown $\mathbf{b}_p$ is exact as well:

$$\mathbf{r}_p = a_p\,\mathbf{d}_p + (1 - a_p)\,\mathbf{e}_p + \mathbf{n}_p, \qquad \mathbf{d}_p := \mathbf{t} - \hat{\mathbf{b}}_p. \tag{2}$$

(Expand $a\mathbf{t} + (1-a)\mathbf{b} - \hat{\mathbf{b}} = a(\mathbf{t} - \hat{\mathbf{b}}) + (1-a)(\mathbf{b} - \hat{\mathbf{b}})$.) Equation (2) is the one to compute with, since $\mathbf{d}_p$ is known once a library spectrum is chosen. On background pixels $\mathbf{r} = \mathbf{e} + \mathbf{n}$, so $\Sigma_r = \Sigma_e + \Sigma_n$.

Remark 2.2 (Clutter-limited, not noise-limited). EnMAP's stated SNR is 354 (VNIR) and 260 (SWIR) (LLD section 4). At reflectance 0.15 that is a noise standard deviation of $4\text{–}6\times10^{-4}$, or 0.005–0.008 normalised units[C]. The measured hidden-pixel background error of FT04 is 0.057–0.139 mean absolute normalised units per band on the four PRISMA bench scenes (`decoder-capacity/README.md`, table 2)[M]. So $\Sigma_n \ll \Sigma_e$ by one to two orders of magnitude, and every detector below is limited by how well the model predicts the background, not by photon noise.

2.2 Residual Mahalanobis as a generalised likelihood ratio

RM (`residual_mahalanobis.py`) takes the eigendecomposition $\hat{\Sigma}_r = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$ of the residual covariance over up to 200,000 valid pixels (fixed seed), keeps the top $d = 30$ components, and whitens:

$$\mathbf{W} = \mathbf{\Lambda}_d^{-1/2}\mathbf{U}_d^\top \in \mathbb{R}^{d\times N}, \qquad \mathrm{RM}(p) = \left\|\mathbf{W}\left(\mathbf{r}_p - \bar{\mathbf{r}}_{\mathscr{R}(p)}\right)\right\|^2,$$

where $\bar{\mathbf{r}}_{\mathscr{R}(p)}$ is the mean over a ring, the 61×61 window minus the central 21×21 (the PRISMA setting), falling back to the global mean when the ring holds fewer than $2d = 60$ valid pixels. (The code subtracts the ring mean in whitened coordinates; since $\mathbf{W}$ is linear the two are the same.)

Proposition 2.3 (RM is the GLRT for an unrestricted signal)**.** *Suppose* $\mathbf{W}\mathbf{r}_p \sim \mathcal{N}(\mathbf{W}\mathbf{s} + \mathbf{W}\mathbf{m}, \mathbf{I}_d)$ *with known local mean* $\mathbf{m}$*, and test* $H_0 : \mathbf{s} = 0$ *against* $H_1 : \mathbf{s} \in \mathbb{R}^N$ *arbitrary. Then* $2\log\Lambda = \mathrm{RM}(p)$*, distributed* χ^2_d *under* H_0*.*

Proof. Under H_1 the maximum of the likelihood over $\mathbf{s}$ is attained at any $\mathbf{s}$ with $\mathbf{W}\mathbf{s} = \mathbf{W}(\mathbf{r}_p - \mathbf{m})$, which makes the exponent zero. Under H_0 the exponent is $-\frac{1}{2}\|\mathbf{W}(\mathbf{r}_p - \mathbf{m})\|^2$. The difference of log-likelihoods, doubled, is RM. Its law under H_0 is that of a sum of d squared standard normals. □

RM therefore asks "is the residual unusual in *any* direction of the leading residual subspace?". Two properties of this choice matter for matching.

Remark 2.4 (What RM does not see). RM projects onto the $d = 30$ *largest*-variance residual directions before whitening. Within them, the low-variance directions are amplified, which is the sense in which it "rewards errors in directions the background never takes" (its docstring). But a residual component in $\mathrm{range}(\mathbf{U}_d)^\perp$, the 135 or more smallest-variance directions on the 165-band grid, contributes nothing. That is a deliberate regularisation (full-band RX was near-singular and was dropped in 2026-05), and it is harmless for detection when targets share the leading residual directions. For identification it may not be: Section 4 shows that the parts of library spectra that separate families carry a few per cent of their energy, and nothing guarantees that part lies inside $\mathrm{range}(\mathbf{U}_d)$. A family test should use its own whitening (as `family_match.py` does, full band with a ridge) rather than inherit RM's truncation.

Proposition 2.5 (The price of not knowing the direction)**.** *If the signal is known to lie in a* q*-dimensional subspace* $\mathrm{range}(\mathbf{H})$*, the GLRT is* $\|\mathbf{P}_{\mathbf{WH}}\mathbf{W}\mathbf{r}\|^2 \sim \chi^2_q(\delta)$ *with noncentrality* $\delta = \|\mathbf{W}\mathbf{s}\|^2$*. At* $P_{\mathrm{FA}} = 10^{-3}$ *and* $P_D = 0.5$ *the required noncentrality is*[C]

dimension q	*1*	*3*	*10*	*30 (RM)*
χ^2_q *threshold*	*10.83*	*16.27*	*29.59*	*59.70*
required δ	*10.83*	*14.24*	*20.53*	*30.59*

Proof. Same argument as Proposition 2.3 with $\mathbf{s} = \mathbf{H}\boldsymbol{\theta}$: the maximiser projects $\mathbf{W}\mathbf{r}$ onto $\mathrm{range}(\mathbf{WH})$. The table is χ^2 and noncentral-χ^2 quantiles. □

Restricting from "any of 30 directions" to a rank-3 family subspace lowers the required SNR by a factor $30.6/14.2 \approx 2.1$, a factor 1.47 in fill, under the Gaussian model. Family matching is RM restricted to library directions; the restriction buys sensitivity and, more importantly, a name.

3 Known-target detection on the residual

Fix one candidate target spectrum $\mathbf{t}$, so $\mathbf{d} = \mathbf{t} - \hat{\mathbf{b}}_p$ is known. From (2),

$$\mathbf{r} = a\,\mathbf{d} + \mathbf{w}, \qquad \mathbf{w} = (1-a)\mathbf{e} + \mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{S}), \tag{3}$$

with $\mathbf{S} = (1-a)^2\Sigma_e + \Sigma_n \approx \Sigma_r$ for the small fills of interest. Write $\langle\mathbf{u},\mathbf{v}\rangle_{\mathbf{S}} = \mathbf{u}^\top\mathbf{S}^{-1}\mathbf{v}$ and $\|\mathbf{u}\|^2_{\mathbf{S}} = \langle\mathbf{u},\mathbf{u}\rangle_{\mathbf{S}}$.

3.1 Fill estimate and its bound

Proposition 3.1 (Fill estimate and Cramér-Rao bound)**.** *The maximum-likelihood estimate of a in (3) is*

$$\hat{a} = \frac{\langle \mathbf{d}, \mathbf{r} \rangle_{\mathbf{S}}}{\|\mathbf{d}\|_{\mathbf{S}}^2}, \qquad \mathbb{E}\hat{a} = a, \qquad \operatorname{Var} \hat{a} = \frac{1}{\|\mathbf{d}\|_{\mathbf{S}}^2},$$

and $\operatorname{Var} \hat{a}$ *equals the Cramér-Rao bound, so $\hat{a}$ is efficient.*

Proof. The log-likelihood is $-\frac{1}{2}\|\mathbf{r} - a\mathbf{d}\|_{\mathbf{S}}^2$ up to a constant, a quadratic in a minimised at $\hat{a}$. Linearity gives $\mathbb{E}\hat{a} = a$ and $\operatorname{Var} \hat{a} = \mathbf{d}^\top \mathbf{S}^{-1} \mathbf{S} \mathbf{S}^{-1} \mathbf{d} / \|\mathbf{d}\|_{\mathbf{S}}^4$. The Fisher information is $\|\mathbf{d}\|_{\mathbf{S}}^2$, so the bound is attained. □

Proposition 3.2 (The $0 \le a \le 1$ constraint)**.** *The constrained MLE is $\tilde{a} = \min(1, \max(0, \hat{a}))$, and $|\tilde{a} - a| \le |\hat{a} - a|$ for every realisation, so its mean squared error is never larger. It is biased upward near $a = 0$: $\mathbb{E}\tilde{a} - a = \mathcal{I}^{-1/2}\big[\phi_{\mathcal{N}}(\kappa) - \kappa\, \Phi(-\kappa)\big]$ with Fisher information $\mathcal{I} = \|\mathbf{d}\|_{\mathbf{S}}^2$, $\kappa = a\sqrt{\mathcal{I}}$, $\phi_{\mathcal{N}}$ and Φ the standard normal density and distribution function, ignoring the upper clip.*

Proof. The likelihood is a concave quadratic in a, so on an interval its maximiser is the projection of the unconstrained one. Projection onto a convex set containing a is non-expansive. The bias is $\mathbb{E}\max(0, \hat{a}) - a$ for $\hat{a} \sim \mathcal{N}(a, 1/\mathcal{I})$: $\mathbb{E}\max(0, \hat{a}) = a\Phi(\kappa) + \mathcal{I}^{-1/2}\phi_{\mathcal{N}}(\kappa)$. □

Proposition 3.3 (Bias from a wrong signature)**.** *If the true target is $\mathbf{t}' = \mathbf{t} + \boldsymbol{\delta}_t$ and the estimator uses $\mathbf{t}$, then*

$$\mathbb{E}\hat{a} = a\Big(1 + \frac{\langle \mathbf{d}, \boldsymbol{\delta}_t \rangle_{\mathbf{S}}}{\|\mathbf{d}\|_{\mathbf{S}}^2}\Big),$$

and the noise-free cosine between $\mathbf{r}$ and $\mathbf{d}$ in the $\mathbf{S}^{-1}$ metric is $\langle \mathbf{d}, \mathbf{d} + \boldsymbol{\delta}_t \rangle_{\mathbf{S}} / (\|\mathbf{d}\|_{\mathbf{S}} \|\mathbf{d} + \boldsymbol{\delta}_t\|_{\mathbf{S}})$.

Proof. Substitute $\mathbf{r} = a(\mathbf{d} + \boldsymbol{\delta}_t) + \mathbf{w}$ into $\hat{a}$. □

A mismatch orthogonal to $\mathbf{d}$ in the $\mathbf{S}^{-1}$ metric does not bias the fill but lowers the cosine; a mismatch along $\mathbf{d}$ is indistinguishable from a different fill. This is the formal content of the LLD's finding that the finite-target matched filter's fill estimate is "unbiased and at the theoretical limit on synthetic data" (Proposition 3.1) yet "collapsed towards 0 on the Gudalur sheets and underestimated oil by about 0.8" on real materials[M]: there the library spectrum is not the deployed material, and $\langle \mathbf{d}, \boldsymbol{\delta}_t \rangle_{\mathbf{S}} < 0$.

3.2 Detectors: matched filter, ACE, and the GLRT for unknown fill and scale

Proposition 3.4 (Matched filter)**.** $T_{\mathrm{MF}} = \langle \mathbf{d}, \mathbf{r} \rangle_{\mathbf{S}} / \|\mathbf{d}\|_{\mathbf{S}} \sim \mathcal{N}(a\|\mathbf{d}\|_{\mathbf{S}}, 1)$. *The detection SNR is* $\mathrm{SNR} = a^2 \|\mathbf{d}\|_{\mathbf{S}}^2 = a^2\, \mathbf{d}^\top \mathbf{S}^{-1} \mathbf{d}$*, and*

$$P_D = Q\big(Q^{-1}(P_{\mathrm{FA}}) - a\|\mathbf{d}\|_{\mathbf{S}}\big), \qquad a_{\min} = \frac{Q^{-1}(P_{\mathrm{FA}}) - Q^{-1}(P_D)}{\|\mathbf{d}\|_{\mathbf{S}}}.$$

For one-sided $a \ge 0$ the GLRT is $\max(0, T_{\mathrm{MF}})^2$, which is why the shipped ACE is signed.

When $\mathbf{S}$ is replaced by a sample covariance from K background pixels the statistic is Kelly's GLRT or the adaptive matched filter of Robey et al.; with $K = 2 \times 10^5$ samples and $N \le 372$ the estimation loss (a factor of roughly $(K - N)/K$ in SNR) is negligible.

Proposition 3.5 (ACE is the GLRT for unknown fill and unknown scale)**.** *If $\mathbf{S} = \sigma^2 \mathbf{S}_0$ with $\mathbf{S}_0$ known and σ^2 unknown under both hypotheses, the GLRT for $H_0 : a = 0$ versus $H_1 : a \ne 0$ reduces to*

$$T_{\mathrm{ACE}} = \frac{(\mathbf{d}^\top \mathbf{S}_0^{-1} \mathbf{r})^2}{(\mathbf{d}^\top \mathbf{S}_0^{-1} \mathbf{d})(\mathbf{r}^\top \mathbf{S}_0^{-1} \mathbf{r})} = \cos^2 \angle\big(\mathbf{S}_0^{-1/2} \mathbf{d}, \mathbf{S}_0^{-1/2} \mathbf{r}\big),$$

which is invariant to σ (CFAR), and $T_{\mathrm{ACE}} \sim \mathrm{Beta}(\frac{1}{2}, \frac{N-1}{2})$ under a Gaussian H_0 [9].

Proof. Whiten by $\mathbf{S}_0^{-1/2}$; write $\tilde{\mathbf{r}}, \tilde{\mathbf{d}}$. Under H_1, maximising over a leaves the residual $\|\mathbf{P}_{\tilde{\mathbf{d}}}^{\perp}\tilde{\mathbf{r}}\|^2$ and maximising over σ^2 gives $\hat{\sigma}_1^2 = \|\mathbf{P}^{\perp}\tilde{\mathbf{r}}\|^2/N$; under H_0, $\hat{\sigma}_0^2 = \|\tilde{\mathbf{r}}\|^2/N$. The GLR is $(\hat{\sigma}_0^2/\hat{\sigma}_1^2)^{N/2}$, a monotone function of $\|\mathbf{P}_{\tilde{\mathbf{d}}}\tilde{\mathbf{r}}\|^2/\|\tilde{\mathbf{r}}\|^2$. Under a white Gaussian H_0 the squared cosine of a spherically symmetric vector with a fixed axis is Beta$(\frac{1}{2}, \frac{N-1}{2})$. □

The shipped matcher computes ACE against $\mathbf{t} - \mathbf{m}$ with $\mathbf{m}$ the 13×13-minus-5×5 ring mean and $\mathbf{S}_0$ the whole-scene pixel covariance with a ridge of 10^{-3} times its mean eigenvalue (`family_match.py`, `whitener`, `ace`). It is a classical detector: no learned background. The residual-domain version replaces $\mathbf{m}$ by $\hat{\mathbf{b}}_p$ and the scene covariance by Σ_r. Section 5 is about what that substitution costs.

3.3 Worked numbers: what fill is detectable?

Worked example 3.6 (A metal roof over grass)**.** Take $\mathbf{t}$ = galvanised sheet metal (USGS GDS334) and $\mathbf{b} = 0.7\times$cheatgrass $+0.3\times$ an alfisol soil, both from the shipped library on the 165-band grid. Then $\overline{|\mathbf{t} - \mathbf{b}|} = 0.170$ reflectance per band, 2.18 normalised units, and $\|\mathbf{D}^{-1}(\mathbf{t} - \mathbf{b})\|^2 = 904^{[C]}$.

The residual covariance is not isotropic: RM exists because "a scene's normal residuals are large along a few shared directions". I use a two-level model[A],

$$\mathbf{S} = \sigma_e^2\Big[(1-\varepsilon)\tfrac{N}{k}\,\mathbf{U}_k\mathbf{U}_k^{\top} + \varepsilon\tfrac{N}{N-k}\big(\mathbf{I} - \mathbf{U}_k\mathbf{U}_k^{\top}\big)\Big], \qquad \operatorname{tr}\mathbf{S} = N\sigma_e^2,$$

with a fraction $1-\varepsilon$ of the error energy in $k = 10$ shared directions. If a fraction φ of $\|\mathbf{d}\|^2$ lies outside range$(\mathbf{U}_k)$ (its *foreignness*), then $\|\mathbf{d}\|_{\mathbf{S}}^2 = \|\mathbf{d}\|^2\big[(1-\varphi)/\lambda_{\text{in}} + \varphi/\lambda_{\text{out}}\big]$ with $\lambda_{\text{in}} = (1-\varepsilon)N\sigma_e^2/k$, $\lambda_{\text{out}} = \varepsilon N\sigma_e^2/(N-k)$. For a Gaussian error, mean $|e| = \sigma_e\sqrt{2/\pi}$, so the measured mean absolute errors 0.05, 0.10, 0.14 give $\sigma_e = 0.063, 0.125, 0.175$. The foreignness of real objects' difference from their ring, relative to the scene's top-10 spectral subspace, has median 0.006 (comment in `implant.py`, bench-v2 diagnosis)[M]; for library families against the top-10 subspace of the library's natural spectra it is 0.000–0.024[C], in agreement.

Table 1: Minimum detectable fill $a_{\min}$ (Proposition 3.4, $P_{\text{FA}} = 10^{-3}$, $P_D = 0.5$, required SNR 9.55) for the metal-over-grass difference of Worked example 3.6, two-level covariance with $\varepsilon = 0.1$, $k = 10$[C]. Measured inputs[M]: background error level; $\varphi = 0.006$.

mean $\lvert e\rvert$ (normalised)	$\varphi = 0.006$	$\varphi = 0.05$	$\varphi = 0.2$	$\varphi = 1$
0.05 (set1-like)	0.018	0.009	0.005	0.002
0.10	0.037	0.018	0.009	0.004
0.14 (set2/set4-like)	0.051	0.025	0.013	0.006

The reality check. The bench numbers let us estimate the signal real targets carry. FT04's hidden-pixel error is 0.18–0.22 on target pixels against 0.06–0.14 on background (`decoder-capacity/README.md`)[M]. Treating the two as adding in quadrature (exact for root-mean-square of independent parts, heuristic for mean absolute values), the target part is $\sqrt{0.218^2 - 0.057^2} = 0.21$ (set1), 0.17 (set2), 0.12 (set3), 0.13 (set4) per band. Divided by the metal example's 2.18, the average bench target is "worth" a metal-over-grass fill of 0.06–0.10. Even at $\varphi = 0.006$, Table 1 then predicts $P_D \geq 0.75$ at $P_{\text{FA}} = 10^{-3}$ in every scene (set3, the worst: $a \approx 0.057$ against $a_{\min} \approx 0.047$ at mean $|e| = 0.13$) and near-certain detection on set1. The measurement says otherwise: Heavy (FTH3 epoch 5) catches 16.0% of objects at the top 0.1% and 49.4% at the top 1%, $\text{pAUC}_{\leq 1\%}$ 0.189 (`FTH3-heavy-joint-more2/README.md`)[M]. The Gaussian $a_{\min}$ is a lower bound that is wrong by an order of magnitude, for identifiable reasons:

1. **Tails.** The null of the statistic is heavy-tailed (RM is "chi-squared-like, heavy right tail"). With a unit-variance Student-t_3 null the 10^{-3} threshold moves from 3.09 to 5.90, a factor 3.6 in required SNR; t_5 gives 2.2[C].

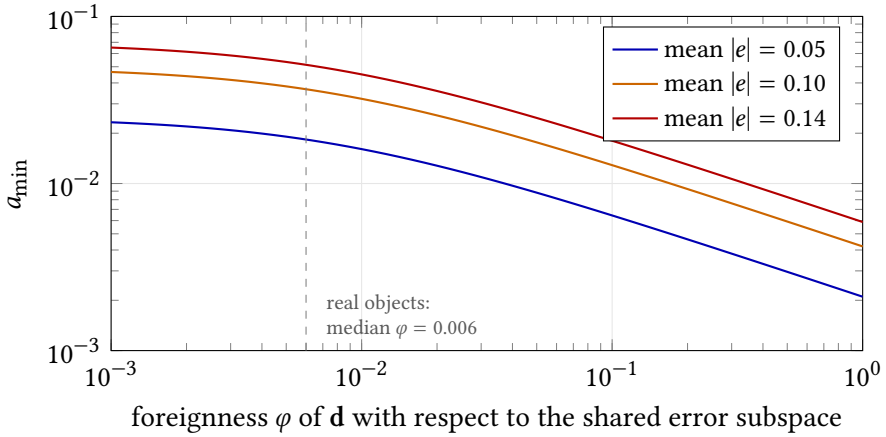


Figure 1: Gaussian-theory minimum detectable fill against foreignness (Table 1). Real objects sit at the far left, where the target hides in the same directions as the model's own error.

2. **Non-stationarity.** The error is largest exactly where targets are: FT06's predicted error scale sits at the 77th–90th percentile of the scene on target pixels (`FT06-uncertainty/b_at_targets_output.txt`)[M]. A global S understates local clutter there.
3. **Alignment.** Real objects have $\varphi \approx 0.006$; most of their signal lies in the directions the error already occupies (Figure 1, left edge).
4. **Leakage and spread.** The model partly reconstructs the target (Section 5) and the PSF spreads it (Worked example 1.4).

The practical conclusion: detectability must be calibrated empirically, by implants at controlled fill (bench-v2), and per-scene percentile thresholds (Section 6) are a necessity rather than a convenience.

For the fill estimate itself, Proposition 3.1 gives $\mathrm{sd}(\hat{a}) = 1/\|\mathbf{d}\|_{\mathbf{S}} = 0.012$ at mean $|e| = 0.10$, $\varphi = 0.006$[C]: precise in principle. The real failures of fill estimation are bias (Proposition 3.3), not variance.

4 Families as subspaces

4.1 How low-rank is a family?

Let $\mathbf{H}_k^{\mathrm{lib}} = [\mathbf{t}_{k1}, \ldots, \mathbf{t}_{km_k}]$ hold family k's resampled spectra in normalised units. On the 165-band grid each man-made family is nearly rank one: the leading singular direction holds 0.89 (textile) to 0.99 (wood/paper) of the energy, and 2–4 directions hold 99%[C] (Table 2). A family is "one shape at varying brightness, plus a little". That is good for detection, and it is also the root of the confusion problem: the shapes of different families are close.

Table 2: Man-made families of the shipped library on the 165-band grid, normalised units[C]. Rank_{99}: number of singular directions holding 99% of the energy (uncentred).

family	m_k	top-1 share	rank_{99}	family	m_k	top-1 share	rank_{99}
building	11	0.940	2	paint	8	0.965	2
glass	1	1.000	1	pavement	15	0.977	2
metal	13	0.949	2	plastic	38	0.930	4
oil	30	0.947	3	roofing	27	0.917	4
textile	14	0.893	4	wood/paper	7	0.987	2

m_k counts entries finite on all 165 bands (oil 30 of 40, plastic 38 of 39, wood/paper 7 of 8).

4.2 Simplex ACE is ACE over the family's convex hull

The shipped man-made family score is the squared norm of the projection of the whitened difference $\mathbf{z} = \mathbf{W}(\mathbf{x} - \mathbf{m})$ onto the cone $\{\sum_j \alpha_j \mathbf{W}(\mathbf{t}_{kj} - \mathbf{m}) : \alpha \geq 0\}$, solved by NNLS, divided by $\|\mathbf{z}\|^2$ (`family_scores`).

Proposition 4.1 (The cone of differences)**.** $\left\{\sum_j \alpha_j(\mathbf{t}_{kj} - \mathbf{m}) : \alpha_j \geq 0\right\} = \left\{A(\bar{\mathbf{t}} - \mathbf{m}) : A \geq 0,\ \bar{\mathbf{t}} \in \mathrm{conv}\{\mathbf{t}_{kj}\}\right\}$. *Hence Simplex ACE equals* $\max_{\bar{\mathbf{t}} \in \mathrm{conv}} \max_{A \geq 0}$ *of the signed ACE against* $\bar{\mathbf{t}}$*, and the NNLS mass* $A = \sum_j \alpha_j$ *is a fill estimate for the best convex mixture.*

Proof. For $A = \sum \alpha_j > 0$ set $\bar{\mathbf{t}} = \sum(\alpha_j/A)\mathbf{t}_{kj}$; then $\sum_j \alpha_j(\mathbf{t}_{kj} - \mathbf{m}) = A(\bar{\mathbf{t}} - \mathbf{m})$. Conversely any $A(\bar{\mathbf{t}} - \mathbf{m})$ with convex weights c_j is $\sum(Ac_j)(\mathbf{t}_{kj} - \mathbf{m})$. For a closed convex cone $\mathscr{K}$, $\|\Pi_{\mathscr{K}}\mathbf{z}\|^2/\|\mathbf{z}\|^2 = \max_{\mathbf{v} \in \mathscr{K}} \cos^2_+(\mathbf{z}, \mathbf{v})$ (Moreau), and every ray of $\mathscr{K}$ is one $\bar{\mathbf{t}}$. □

So Simplex ACE lets the deployed material be any convex mixture of the family's lab spectra, at any non-negative fill; it can never score below the best single member (the LLD's unit test). It cannot represent a brightness change of the target *relative to* its mixture weights unless the family contains spectra at several brightnesses, which Table 2 says it effectively does.

4.3 The matched subspace detector with nuisance subspaces

Proposition 4.2 (Matched subspace detector with interference [20])**.** *Let* $\mathbf{r} = \mathbf{H}\boldsymbol{\theta} + \mathbf{J}\boldsymbol{\psi} + \mathbf{w}$ *with* $\mathbf{w} \sim \mathcal{N}(0, \sigma^2\mathbf{S}_0)$*, signal subspace* $\mathbf{H}$ *(rank* q*), interference subspace* $\mathbf{J}$ *(rank* j*), both unknown in amplitude. Whitening by* $\mathbf{S}_0^{-1/2}$ *(tildes) and writing* $\mathbf{P}^{\perp}_{\tilde{\mathbf{J}}}$ *for the projection away from* $\tilde{\mathbf{J}}$*,* $\mathbf{H}_{\perp} = \mathbf{P}^{\perp}_{\tilde{\mathbf{J}}}\tilde{\mathbf{H}}$*, the GLRT is*

$$T_{\mathrm{MSD}} = \frac{\|\mathbf{P}_{\mathbf{H}_{\perp}}\mathbf{P}^{\perp}_{\tilde{\mathbf{J}}}\tilde{\mathbf{r}}\|^2}{\|\mathbf{P}^{\perp}_{[\tilde{\mathbf{H}}\ \tilde{\mathbf{J}}]}\tilde{\mathbf{r}}\|^2} \cdot \frac{N - q - j}{q} \ \sim\ F_{q,\,N-q-j} \quad \text{under } H_0,$$

and with σ *known the numerator alone, divided by* σ^2*, is* χ^2_q*. The scale-invariant form normalised by* $\|\mathbf{P}^{\perp}_{\tilde{\mathbf{J}}}\tilde{\mathbf{r}}\|^2$ *instead is the adaptive subspace detector [10], the subspace ACE.*

Proposition 4.3 (Illumination gain and offset are a known interference subspace)**.** *Let the observed target be* $g\mathbf{t} + c\mathbf{1}$ *(gain from slope, shadow or BRDF; offset from adjacency or haze), so* $\mathbf{x} = a(g\mathbf{t} + c\mathbf{1}) + (1 - a)\mathbf{b} + \mathbf{n}$*. Then, exactly,*

$$\mathbf{x} - \hat{\mathbf{b}} = ag\,(\mathbf{t} - \hat{\mathbf{b}}) + a(g-1)\,\hat{\mathbf{b}} + ac\,\mathbf{1} + (1-a)\mathbf{e} + \mathbf{n}.$$

The nuisance directions $\hat{\mathbf{b}}$ *and* $\mathbf{1}$ *are known at each pixel, so* $\mathbf{J} = [\hat{\mathbf{b}}, \mathbf{1}]$ *(mapped by* $\mathbf{D}^{-1}$ *in normalised units) and Proposition 4.2 applies. Only the product* ag *is identifiable; fill and target brightness are confounded.*

Proof. Expand: $ag\mathbf{t} - ag\hat{\mathbf{b}} + ag\hat{\mathbf{b}} - a\hat{\mathbf{b}} + ac\mathbf{1} + (1-a)\mathbf{b} - \hat{\mathbf{b}} + a\hat{\mathbf{b}} = ag\mathbf{t} + ac\mathbf{1} + (1-a)\mathbf{b} - \hat{\mathbf{b}}$, which is $\mathbf{x} - \hat{\mathbf{b}}$ less $\mathbf{n}$. Scaling $g \mapsto \gamma g$, $a \mapsto a/\gamma$ leaves ag and changes only the interference amplitude $a(g-1)$, which is a free parameter. □

Projecting out $\hat{\mathbf{b}}$ costs one degree of freedom and removes the target's component along its own background, which for a dark target on a bright background is most of the target. This is the trade every brightness-robust detector makes, made explicit; the cheaper alternative is to rely, as Simplex ACE does, on the family containing several brightnesses.

4.4 Energy decomposition and the projection angle

In whitened coordinates $\tilde{\mathbf{r}}$, for family subspace $\tilde{\mathbf{H}}_k$,

$$\|\tilde{\mathbf{r}}\|^2 = E_k^{\text{expl}} + E_k^{\text{unex}}, \quad E_k^{\text{expl}} = \|\mathbf{P}_{\tilde{\mathbf{H}}_k}\tilde{\mathbf{r}}\|^2, \quad E_k^{\text{unex}} = \|\mathbf{P}_{\tilde{\mathbf{H}}_k}^{\perp}\tilde{\mathbf{r}}\|^2, \qquad \cos^2\theta_k = \frac{E_k^{\text{expl}}}{\|\tilde{\mathbf{r}}\|^2} = \text{ACE}_k,$$

the family-explained and unexplained energies and the projection angle θ_k. RM is the total whitened energy in 30 leading directions; the family score is the share explained by $q_k \le 4$ directions; the unexplained energy is the lack-of-fit used for abstention in Section 6. Under a white Gaussian null the squared cosine to a fixed q-dimensional subspace is $\text{Beta}(q/2, (N-q)/2)$, mean q/N: on 165 bands, 0.006 for $q = 1$ and 0.018 for $q = 3$, with 99.5th percentiles 0.047 and 0.076[C]. A family of larger effective rank collects more null energy, which is why raw scores of different families are not comparable.

4.5 Why SAM fails on mixed pixels

Proposition 4.4 (SAM on a two-component mixture)**.** *Let* $\mathbf{x} = a\mathbf{t} + (1-a)\mathbf{b}$ *with* $\theta_{bt} = \angle(\mathbf{b}, \mathbf{t})$. *Then* $\mathbf{x}$ *lies in* $\text{span}\{\mathbf{b}, \mathbf{t}\}$ *and*

$$\tan\angle(\mathbf{x}, \mathbf{b}) = \frac{a\|\mathbf{t}\|\sin\theta_{bt}}{(1-a)\|\mathbf{b}\| + a\|\mathbf{t}\|\cos\theta_{bt}}, \qquad \angle(\mathbf{x}, \mathbf{t}) = \theta_{bt} - \angle(\mathbf{x}, \mathbf{b}).$$

For small a, $\angle(\mathbf{x}, \mathbf{b}) \approx a\,(\|\mathbf{t}\|/\|\mathbf{b}\|)\sin\theta_{bt}$*: the pixel stays within a few degrees of its background, so any library entry close to the background beats the target.*

Proof. Place $\mathbf{b}$ on the first axis of the plane; $\mathbf{t}$ has coordinates $\|\mathbf{t}\|(\cos\theta_{bt}, \sin\theta_{bt})$. Take the coordinates of $\mathbf{x}$ and read off the angle. □

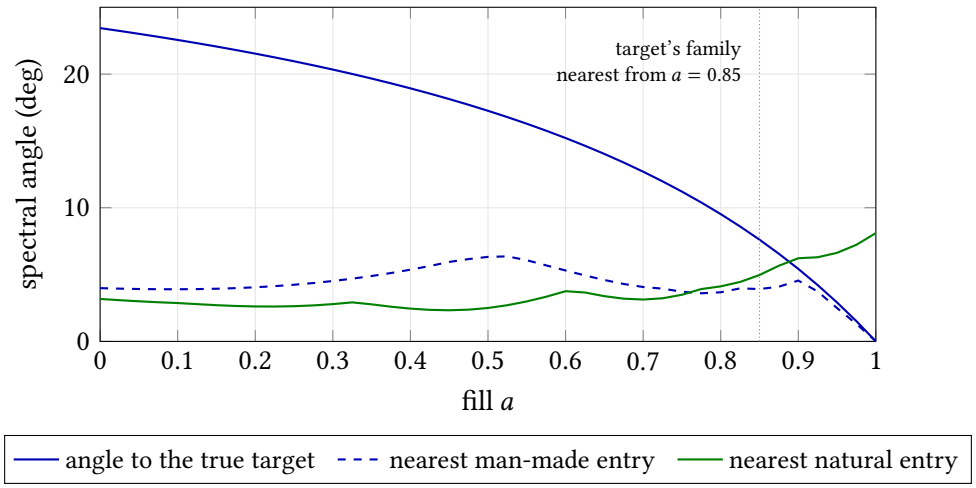


Figure 2: SAM on galvanised metal mixed into a grass/soil background, against all 352 complete library entries on the 165-band grid[C]. The nearest entry is non-photosynthetic vegetation (2.5–2.9°) for every fill up to 0.5; some man-made entry is nearest from $a = 0.775$, the target's own family only from $a = 0.85$. The LLD measured 6% right at 30% fill on the planted set[M].

Subtracting a background estimate removes the problem at its source: $\mathbf{x} - \hat{\mathbf{b}} = a(\mathbf{t} - \hat{\mathbf{b}}) + (1-a)\mathbf{e} + \mathbf{n}$ points along the target difference whatever a is, up to error. That is the case for matching on the residual, and for the ring-mean subtraction the shipped matcher already does.

Table 3: Smallest principal angle between rank-3 family subspaces in normalised units[C], raw, and after projecting out a 10-dimensional "background" subspace (top-10 principal directions of the library's centred natural spectra, the analogue of `implant.background_subspace`).

pair	raw	after removing background-10
metal vs roofing	3.70°	49.8°
metal vs plastic	2.04°	43.2°
plastic vs textile	4.53°	32.7°
paint vs plastic	5.92°	19.4°
pavement vs building	0.51°	10.0°

4.6 How far apart are families?

Two readings of Table 3. Raw family subspaces are a few degrees apart; the fraction of a unit target's energy that separates two families is $\sin^2$ of the angle, 8×10^{-5} (pavement/building) to 0.011 (paint/plastic). Once the directions natural surfaces share are removed, the families are well separated, but that part carries only 0.0–2.4% of a family's difference from the natural mean (its foreignness, Section 3)[C]. Identification lives in a few per cent of the signal energy. Every error source in Section 7 has to be compared with that few per cent, not with the whole signal.

5 Leakage: when the background prediction partly reconstructs the target

5.1 Scalar leakage

Model 5.1 (Proportional leakage)**.** $\hat{\mathbf{b}}_p = \mathbf{b}_p + l\,a_p(\mathbf{t}-\mathbf{b}_p) - \mathbf{e}_p$ with leakage fraction $l \in [0,1]$.

Proposition 5.2. *Under Model 5.1,* $\mathbf{r}_p = a_p(1-l)(\mathbf{t}-\mathbf{b}_p)+\mathbf{e}_p+\mathbf{n}_p$ *and* $\mathbf{d}_p = \mathbf{t}-\hat{\mathbf{b}}_p = (1-la_p)(\mathbf{t}-\mathbf{b}_p)+\mathbf{e}_p$. *Hence (i) the detection SNR is multiplied by* $(1-l)^2$ *and* $a_{\min}$ *by* $1/(1-l)$*; (ii) in the noise-free limit* $\hat{a} \to a(1-l)/(1-la)$*, a downward bias; (iii) the direction of the residual is unchanged, so scale-invariant statistics (ACE, Simplex ACE, family percentiles) lose only through SNR.*

Proof. Substitute into $\mathbf{r} = \mathbf{x} - \hat{\mathbf{b}}$ and into Proposition 3.1 with $\mathbf{e} = \mathbf{n} = 0$: $\hat{a} = a(1-l)\|\mathbf{u}\|^2/((1-la)\|\mathbf{u}\|^2)$ for $\mathbf{u} = \mathbf{t} - \mathbf{b}$. □

For $l = 0.1, 0.3, 0.5$ the SNR falls to $0.81, 0.49, 0.25$ of its leak-free value[C] (Figure 3).

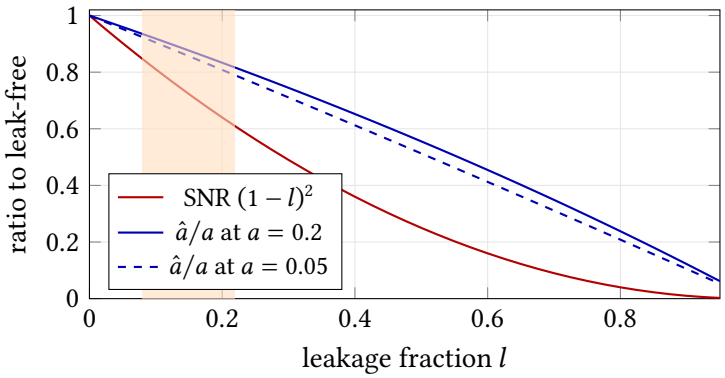


Figure 3: Cost of proportional leakage (Proposition 5.2). The band is the linear-interpolator estimate for a one-pixel target and 4-px tokens (Worked example 5.5)[C].

5.2 Operator leakage and a leakage-free subspace

The learned predictor need not leak proportionally. Linearise f_θ around the target-free context: with $\mathbf{J}_{pq} = \partial \hat{\mathbf{b}}_p / \partial \mathbf{x}_q$ and the target's contribution $a_q(\mathbf{t} - \mathbf{b}_q) \approx a_q(\mathbf{t} - \mathbf{b}_p)$ at visible neighbours $q \in C(p)$,

$$\hat{\mathbf{b}}_p \approx \hat{\mathbf{b}}_p^{(0)} + a_p \mathbf{L}_p(\mathbf{t} - \mathbf{b}_p), \qquad \mathbf{L}_p = \sum_{q \in C(p)} \mathbf{J}_{pq} \frac{a_q}{a_p}. \tag{4}$$

So $\mathbf{r}_p = a_p(\mathbf{I} - \mathbf{L}_p)(\mathbf{t} - \mathbf{b}_p) + \mathbf{e}_p + \mathbf{n}_p$. Since p is hidden, $\mathbf{J}_{pp} = 0$: leakage happens only through the fills a_q at visible neighbours, the PSF halo and any part of the object larger than a token.

Proposition 5.3 (Projected identifiability)**.** *Under* (4)*, families j and k are distinguishable from $\mathbf{r}_p$ only through $(\mathbf{I} - \mathbf{L}_p)\mathbf{H}_j$ and $(\mathbf{I} - \mathbf{L}_p)\mathbf{H}_k$, and the matched statistic must be formed against the leaked library $(\mathbf{I} - \mathbf{L}_p)(\mathbf{t} - \hat{\mathbf{b}}_p)$, not against $\mathbf{t} - \hat{\mathbf{b}}_p$. If $\mathbf{L}_p = \mathbf{P}_{\mathscr{M}}$ is the projection onto a subspace the model can render, the usable signal is $a_p^2 \|\mathbf{P}_{\mathscr{M}}^{\perp}(\mathbf{t} - \mathbf{b})\|_{\mathbf{S}}^2$, and the two families are identifiable iff the smallest principal angle between $\mathbf{P}_{\mathscr{M}}^{\perp}\mathbf{H}_j$ and $\mathbf{P}_{\mathscr{M}}^{\perp}\mathbf{H}_k$ is non-zero.*

Table 3 is the toy version with $\mathscr{M}$ = the natural background subspace: projection makes families *more* separable in angle while removing almost all of their energy. A model that leaked exactly its background subspace would leave a residual in which naming is easier per unit energy and detection is much harder.

Proposition 5.4 (The decoder's range bounds leakage)**.** *For the reduced-attention stem the output layer is linear in a D-dimensional code: $\hat{z}_n = \mathbf{u}(\lambda_n)^\top \boldsymbol{\zeta} + c(\lambda_n)$ in normalised units (*`WavelengthLinearDecoder`*, `spectral_reduced.py`), so $\hat{\mathbf{b}} = \boldsymbol{\mu} + \mathbf{D}(\mathbf{U}\boldsymbol{\zeta} + \mathbf{c})$. Let $\mathbf{Q}$ project onto* range$(\mathbf{DU})^\perp$*. Then $\mathbf{Q}\mathbf{L}_p = 0$ exactly, and*

$$\mathbf{Q}\mathbf{r}_p = \mathbf{Q}\big(\mathbf{x}_p - \boldsymbol{\mu} - \mathbf{D}\mathbf{c}\big),$$

independent of the network. $D = 32$ for FT04 and $D = 64$ for Heavy (`compressed_channels` *in the FT04 and FTH1/FTH3 configs; verified).*

Proof. Any two predictions differ by $\mathbf{DU}(\boldsymbol{\zeta} - \boldsymbol{\zeta}') \in$ range$(\mathbf{DU})$, so the leaked term is in range$(\mathbf{DU})$ and $\mathbf{Q}$ annihilates it. Also $\mathbf{Q}\hat{\mathbf{b}} = \mathbf{Q}(\boldsymbol{\mu} + \mathbf{D}\mathbf{c})$. □

So the residual splits into a part where the model both suppresses background and can leak (range$(\mathbf{DU})$), and a part where it does neither and the "learned" residual is the raw pixel less a fixed offset. FT04's measured floor, the residual energy outside range$(\mathbf{DU})$, is 2–10% on background and 2–10% on targets (`decoder-capacity/README.md`, table 1)[M]: the complement is leak-proof but carries little target. Heavy's floor (with $D = 64$) has not been measured.

5.3 The halo seen through the mask

FT06 trained a head to predict the expected error scale (call it β_p) at hidden pixels. On targets β exceeds its own 3–8 px surroundings by a factor 1.05/1.12/1.04 (epoch 1) and 1.07/1.19/1.13 (epoch 3) on sets 1, 2, 4 (`FT06-uncertainty/README.md`)[M]. A hidden target that the context could not see would give a ratio of one; the excess is direct evidence that $\sum_q \mathbf{J}_{pq} a_q \neq 0$, i.e. leakage through the halo. The geometry is easy to compute.

Worked example 5.5 (Halo exposure and a leakage estimate)**.** A one-pixel target with $\sigma_{\mathrm{PSF}} = 1.4$ px, averaged over its 16 positions in a 4×4 token: 52% of its energy stays in its own (hidden) token and 40% (range 30–47%) lands in the four edge-neighbour tokens, which are visible when its token is hidden. With 8×8-pixel cells (FT08's 2×2-token blocks) the visible share falls to 24% on average, but ranges from 1.6% to 46%: a target near a cell edge leaks as much as before[C]. If the predictor were the mean

of the 16 pixels bordering the hidden token, its leakage would be $l = 0.22$ (range 0.19–0.23); if it were the mean of all 64 pixels of the four neighbour tokens, $l = 0.08$ (0.06–0.10)[C]. A learned predictor sits somewhere in or near that range; by Proposition 5.2 that costs 16–39% of SNR.

Proposition 5.6 (When wider masks pay)**.** *Let cell size c give leakage l_c and background-error variance $\sigma^2_{e,c}$ along the target direction. Moving from c_1 to c_2 raises the detection SNR iff*

$$\left(\frac{1-l_{c_2}}{1-l_{c_1}}\right)^2 > \frac{\sigma^2_{e,c_2}}{\sigma^2_{e,c_1}}.$$

Proof. SNR $= a^2(1-l)^2\|\mathbf{t}-\mathbf{b}\|^2/\sigma_e^2$ along a fixed direction; take the ratio. □

With $l_4 = 0.22$ and $l_8 \approx 0.22 \times 0.24/0.40 = 0.13$ (scaling leakage with halo exposure[A]), the left side is $(0.87/0.78)^2 = 1.24$: FT08 would gain if hiding 8-px blocks raised background error by less than 24%. FT08 measured a resolved loss, pAUC -0.0058 $[-0.0106, -0.0010]$ (`FT07-gated/README.md`)[M], and its README attributes it to more background residual ("normal texture inside a block is rebuilt less exactly"). That is consistent with the inequality failing, but $\sigma_{e,8}/\sigma_{e,4}$ was not measured directly; it is a cheap measurement (Section 10).

Remark 5.7 (The classical ring leaks too). The ring estimate is not leak-free either. Under `psf_weights` the mean weight on the 2–3 px ring that `contrast_z` uses is 0.32 for a 1×1 object, 0.20 for 2×2, 0.12 for 4×4[C]: ring-based contrast and ring-mean matching see $(1-l_{\text{ring}})$ of the signal with l_{ring} of that size. The shipped matcher's 13×13 minus 5×5 ring starts at 3 px, so its leakage is smaller. Learned and ring backgrounds trade leakage against how well they predict $\mathbf{b}_p$.

6 Decision and abstention

6.1 Posterior over families plus "unknown"

Let $k \in \{0, 1, \dots, K\}$ with 0 meaning "none of the library". For a pixel with residual $\mathbf{r}$,

$$P(k \mid \mathbf{r}) \propto \pi_k \int p(\mathbf{r} \mid k, a, \boldsymbol{\theta})\, p(a, \boldsymbol{\theta} \mid k)\, da\, d\boldsymbol{\theta}.$$

A Laplace approximation of the integral gives the profile likelihood (the GLRT statistic) minus a penalty of $\frac{q_k}{2}\log(\cdot)$ for the family's effective dimension. Without that penalty the family with more freedom wins on noise alone; this is the LLD's observation that "uncapped, about 1,900 minerals out-vote the about 180 man-made spectra by sheer number of chances", fixed there by capping each family at 40 spaced entries and by never forming a cone of natural spectra. Chow's rule [1] names $\hat{k} = \arg\max_k P(k \mid \mathbf{r})$ if $\max_k P(k \mid \mathbf{r}) \geq 1 - c$ and abstains otherwise; with an explicit class 0 it abstains by naming 0.

The densities are not known well enough to compute these posteriors, and the scene-to-scene variation of the null is large. The shipped matcher therefore replaces likelihoods by *calibrated* scores.

6.2 Per-scene calibration and the false-name bound

Definition 6.1 (Calibrated family score)**.** For each family k, compute its score T_k on N_0 background pixels sampled from the scene (anomalies grown by 3 px excluded; $N_0 = 10{,}000$; `NULL_N`, `EXCLUDE_BUFFER_PX`). The calibrated score of pixel p is the share of null scores strictly below its own, $\pi_k(p) = \#\{i : T_k(y_i) < T_k(p)\}/N_0$ (`calibrate`). Name the best family if $\max_k \pi_k(p) \geq p^* = 1 - \alpha/F$, with F the number of man-made families.

Proposition 6.2 (False-name rate on ordinary pixels)**.** *If p is exchangeable with the null sample (an ordinary background pixel of the same scene) and scores have no ties, $P(\pi_k(p) \geq p^*) \leq 1 - p^* + 1/N_0$ for each k, and the probability that p receives some man-made name is at most $F(1-p^*) + F/N_0 = \alpha + F/N_0$.*

Proof. Under exchangeability the rank of $T_k(p)$ among the N_0+1 values is uniform on $\{1,\dots,N_0+1\}$. The event $\pi_k \geq p^*$ requires the rank to be among the top $N_0-\lceil p^* N_0\rceil+1$, of probability $(N_0(1-p^*)+1)/(N_0+1)$ at most. Take the union over the F man-made families. □

At the default $\alpha = 5\%$, $F = 10$, $p^* = 0.995$: each family's bar sits at 50 null exceedances, and the realised per-family false-name rate has a sampling standard deviation of 7×10^{-4}, 14% of its target 0.005[C]; with $N_0 = 10^5$ it would be 4%. The LLD measured the realised rate at a quarter to a half of α[M]: the union bound is loose because family scores are strongly correlated (Table 3).

Remark 6.3 (Why the bound says nothing about false alarms). The exchangeability in Proposition 6.2 is with *ordinary* pixels. The pixels the matcher is asked about are anomaly-flagged, unusual by construction, and the bound does not apply: on the four PRISMA scenes 9–24% of false alarms were named man-made at $\alpha = 5\%$ (LLD section 5)[M]. A calibration that does apply is to draw the null from the *candidate* population itself (the top-percentile pixels), which is about 98% non-target (false alarms outnumber targets about 50:1). If a fraction π of that null is target, then for any threshold η, $P_{\text{non-target}}(T > \eta) \leq P_{\text{null}}(T > \eta)/(1-\pi)$, so contamination at $\pi = 0.02$ inflates the guaranteed rate by at most 2%. This conditional calibration controls false names on false alarms, which is the product's actual failure.

6.3 Why a fixed absolute threshold fails

Proposition 6.4. *Let a scene's residual error be $c\,\mathbf{e}_0$ with $\mathbf{e}_0$ from a fixed distribution. A matched-filter threshold η fixed across scenes has false-alarm rate $P(T_{\text{MF}} > \eta/c)$ in the unwhitened case, which moves by orders of magnitude for the scale changes seen across scenes. ACE removes c but not the dependence on the shape of the null (dimension, tail weight, non-stationarity).*

The project rule (`docs/05-detectors.md`, "Thresholding is percentile, never absolute": a calm lake has residuals near 0.01, a fire-affected scene near 0.1) is the same statement for anomaly scores. Family matching inherits it, per family, because the null also depends on the family's effective rank (Section 4).

6.4 Open-set recognition

Naming is open-set: most anomalies are in no library. Three tests, each calibrated, separate the outcomes:

1. *detection*: the pixel is anomalous (RM percentile, upstream);
2. *attribution*: $\max_k \pi_k \geq p^*$ (the shipped rule);
3. *fit*: the unexplained whitened energy $\|\mathbf{P}^{\perp}_{\tilde{\mathbf{H}}_k}\tilde{\mathbf{r}}\|^2$ is not larger than the best family leaves on implanted targets of that family at similar fill (a lack-of-fit test; under a Gaussian H_k it is χ^2_{N-q}-like and the real reference must come from implants);

plus a margin to the runner-up family. A pixel that passes 1 and 2 but fails 3 is "man-made-like, not this family"; one that fails 2 is "man-made, type unconfirmed". This is the open-space-risk idea of [21] with the open space measured by lack of fit. Note one asymmetry in the product: the worker shows natural-family names whether or not they pass p^* (`named = answered or group == "natural"` in `_spectral_library_match_run.py`).

7 Error sources and their scale

Each source enters the statistic through (1) as either a change of target direction (which costs identification) or extra residual energy (which costs detection). The yardstick from Section 4 is the separation

between families: a few degrees, a few per cent of the energy.

7.1 Atmospheric correction residuals

Model the corrected reflectance as $\mathbf{x}^{\text{obs}} = \mathbf{G}\mathbf{x} + \mathbf{o}$ with $\mathbf{G} = \operatorname{diag}(g(\lambda_n))$ and $g, \mathbf{o}$ smooth in wavelength. If the learned background is approximately equivariant[A], $\hat{\mathbf{b}}(\mathbf{G}\mathbf{x}+\mathbf{o}) \approx \mathbf{G}\hat{\mathbf{b}}(\mathbf{x})+\mathbf{o}$ (it is trained on many scenes with different residual corrections), then

$$\mathbf{r}^{\text{obs}} \approx a\,\mathbf{G}(\mathbf{t}-\mathbf{b}) + \mathbf{G}\mathbf{e} + \mathbf{G}\mathbf{n} \;:$$

the offset cancels in the residual (it does not cancel for SAM on raw pixels), and the gain distorts the target direction.

Proposition 7.1 (Angle bound for a gain error)**.** *For positive diagonal* $\mathbf{G}$ *with* $\kappa = g_{\max}/g_{\min}$ *and any* $\mathbf{d} \neq 0$,

$$\cos\angle(\mathbf{G}\mathbf{d}, \mathbf{d}) \;\geq\; \frac{2\sqrt{\kappa}}{1+\kappa}, \qquad 1 - \cos^2\angle(\mathbf{G}\mathbf{d}, \mathbf{d}) \leq \left(\frac{\kappa-1}{\kappa+1}\right)^2.$$

Proof. Put $\mathbf{u} = \mathbf{G}^{1/2}\mathbf{d}$. Then $\mathbf{d}^\top\mathbf{G}\mathbf{d} = \mathbf{u}^\top\mathbf{u}$, $\|\mathbf{G}\mathbf{d}\|^2 = \mathbf{u}^\top\mathbf{G}\mathbf{u}$, $\|\mathbf{d}\|^2 = \mathbf{u}^\top\mathbf{G}^{-1}\mathbf{u}$, and Kantorovich's inequality $(\mathbf{u}^\top\mathbf{G}\mathbf{u})(\mathbf{u}^\top\mathbf{G}^{-1}\mathbf{u}) \leq \frac{(1+\kappa)^2}{4\kappa}\|\mathbf{u}\|^4$ gives $\cos^2 \geq 4\kappa/(1+\kappa)^2$. □

For $\kappa = 1.02, 1.05, 1.1, 1.2$ the worst-case angle is $0.57^\circ, 1.40^\circ, 2.73^\circ, 5.22^\circ$[C]. A random smooth ±5% gain (a cubic Legendre curve squashed into $[0.95, 1.05]$) applied to 200 library-minus-background differences gave a median of 0.61° and a 95th percentile of 1.29°[C], well inside the bound. The bound is Euclidean; in whitened coordinates the low-variance directions are stretched and the angle can be larger.

What size are the gains? The atmospheric-correction work measured land MAE against the vendor L2 product of 0.0146 (EnMAP vs PACO), 0.0241 (PRISMA vs ASI L2C) and 0.0081 (AVIRIS-NG vs JPL), reduced by harmonisation to 0.0094 and 0.0128 for EnMAP and PRISMA (`research/experiments/atm-correction/README.md`)[M]. Against a median reflectance of 0.155 an MAE of 0.015 is about 10%; read as a pure gain error[A] that is $\kappa \approx 1.1$–1.2, a 3–5° worst-case rotation. That is the same size as the raw separation of metal from roofing (3.7°) and larger than metal from plastic (2.0°) or pavement from building (0.5°). **An atmospheric-correction gain error at the level we measure is enough, by itself, to move a target across a family boundary.** These MAEs are processor-against-processor, not against ground truth; since the models were trained on vendor L2, they are the relevant error for deployment from L1.

7.2 Band-set mismatch and resampling

If a band's true centre is $\lambda_n + \Delta$ but the library is resampled at λ_n, the error is $t_n(\Delta) - t_n \approx \Delta\,(s_n * \rho')(\lambda_n)$, bounded by $|\Delta| \max|\rho'|$ because the normalised response integrates to one. Over the 352 complete library entries on the 165-band grid[C]:

perturbation	SAM change, median	SAM change, max	max rel. change, median
centre shift 1 nm	0.11°	0.52°	1.2%
centre shift 2 nm	0.21°	1.04°	2.5%
centre shift 5 nm	0.52°	2.55°	6.1%
FWHM 8 or 12 nm instead of 10	0.03°	0.20°	-
FWHM 15 nm instead of 10	0.06°	0.50°	-

Spectral calibration errors of a nanometre or two cost a tenth to a fifth of a degree for a typical entry: an order of magnitude below the gain error. Width errors are smaller still. Two practical failure modes matter more than accuracy: bands with no FWHM (PRISMA's null FWHM makes the band invalid, LLD gotcha) and per-pixel nodata on a kept band (such a pixel is not scored).

7.3 Library-to-image domain gap

Write the deployed material as $\mathbf{t}' = \mathbf{t} + \boldsymbol{\delta}_t$ (weathering, dust, mixtures within the object, BRDF). Proposition 3.3 gives its effect on fill and cosine. Its size is best read off the one real test with field spectra: on the Gudalur sheets, Simplex ACE names 27% of answered pixels correctly with the library alone and about 78% with the sheets' own field spectra added (LLD section 5)[M], and the absorption-feature check that was right 62% of the time on planted targets was right 0% of the time on the real sheets[M]. Of all error sources this is the largest, and it is the one synthetic tests cannot measure.

7.4 The decoder's output floor

By Proposition 5.4, the part of the background the model can never predict is $\mathbf{Q}(\mathbf{b} - \boldsymbol{\mu} - \mathbf{D}\mathbf{c})$. For FT04 ($D = 32$) it is 2–10% of residual energy on background and on targets; mean absolute floor 0.013–0.033 (background) and 0.022–0.043 (targets), against actual residuals of 0.048–0.122 and 0.149–0.204 (`decoder-capacity/README.md`)[M]. The decoder-capacity study's conclusion follows: over 90% of the error is in which code the decoder predicts, not in what it can render. For matching the floor is a nuisance subspace of known range: project it out (or whiten it separately) and it cannot be mistaken for a family.

7.5 Summary

Table 4: Error sources against the identification yardstick (family separation of 0.5–6° raw).

source	how it enters	size
sensor noise	residual energy	0.005–0.008 norm. units[C]; negligible
background-prediction error	residual energy; sets $\mathbf{S}$	0.05–0.14 mean abs[M]
leakage	SNR $\times(1-l)^2$; direction $\mathbf{I}-\mathbf{L}$	$l \approx$ 0.08–0.22 (estimate)[C]
atmospheric gain	rotation of $\mathbf{t}-\mathbf{b}$	$\leq 2.7°$ at $\kappa = 1.1$; $\kappa \approx$ 1.1–1.2[A]
atmospheric offset	cancels in the residual	(hits SAM on raw pixels)
band centre / width	rotation of $\mathbf{t}$	0.1–0.2° typical at 1–2 nm[C]
library-to-image gap	rotation and bias	dominant: 27% vs 78% right at Gudalur[M]
decoder floor	fixed nuisance subspace	2–10% of residual energy (FT04)[M]

8 The learned namer

8.1 The generator as a sampling distribution

The parked plan (`spectral-matching-fresh-start.md`) trains a namer on synthetic mixtures. Formalise the generator G as a joint law over

$$(k,\ \mathbf{t},\ a,\ \mathbf{x}_C,\ \mathbf{G}, \mathbf{o},\ \mathbf{n}) \sim p_G : \qquad k \sim \pi_G, \qquad \mathbf{t} \sim \text{library}_k \text{ (augmented)},$$
$$a \sim p_G(a) \ \text{(e.g. log-uniform on } [0.05, 0.5]),$$

a real background patch $\mathbf{x}_C$ from the shards, implanted through the PSF (`implant_tile`), a smooth gain and offset, and noise; $k = 0$ is "none" ($a = 0$). The namer sees an input $\boldsymbol{\upsilon} = \Upsilon(\mathbf{x}_p, \mathbf{x}_C)$, for instance $(\mathbf{x}_p, \hat{\mathbf{b}}_p, \mathbf{r}_p)$ in normalised units plus a scene summary of $\boldsymbol{\Sigma}_r$. To simulate leakage faithfully, $\hat{\mathbf{b}}_p$ must be computed by running Heavy on the *implanted* context; feeding a $\hat{\mathbf{b}}$ from the clean patch would teach a leak-free world.

8.2 Objective and its minimiser

Proposition 8.1 (Bayes classifier under the generator)**.** *Over measurable $q(\cdot \mid \boldsymbol{v})$, the expected cross-entropy $\mathbb{E}_G[-\log q(k \mid \boldsymbol{v})]$ is minimised by $q^* = p_G(k \mid \boldsymbol{v})$; the squared-loss fill head is minimised by $\mathbb{E}_G[a \mid \boldsymbol{v}, k \neq 0]$. The classifier $\arg\max_k q^*$ is Bayes-optimal for 0-1 loss under G, and with a reject cost c Chow's rule on q^* is optimal.*

Proof. Conditional on $\boldsymbol{v}$, $\mathbb{E}[-\log q(k \mid \boldsymbol{v})] = H(p_G(\cdot \mid \boldsymbol{v})) + \mathrm{KL}(p_G \| q)$, minimised at $q = p_G$ (Gibbs). The conditional mean minimises squared error. The 0-1 and Chow statements are the standard Bayes and reject-option results. □

Proposition 8.2 (What the input must contain)**.** *Under the Gaussian residual model $\mathbf{r} \mid k, a, \boldsymbol{\theta} \sim \mathcal{N}(a\mathbf{H}_k\boldsymbol{\theta}, \mathrm{S})$ with S known, the statistics $\{\mathbf{H}_k^\top \mathrm{S}^{-1}\mathbf{r}\}_k$ and $\mathbf{r}^\top \mathrm{S}^{-1}\mathbf{r}$ are jointly sufficient for $(k, a, \boldsymbol{\theta})$. A namer given $\mathbf{r}$ without S cannot form them and cannot be calibrated across scenes.*

Proof. The log-likelihood is $-\frac{1}{2}\mathbf{r}^\top \mathrm{S}^{-1}\mathbf{r} + a\boldsymbol{\theta}^\top \mathbf{H}_k^\top \mathrm{S}^{-1}\mathbf{r} - \frac{1}{2}a^2 \boldsymbol{\theta}^\top \mathbf{H}_k^\top \mathrm{S}^{-1}\mathbf{H}_k\boldsymbol{\theta}$; it depends on $\mathbf{r}$ only through the listed statistics (Fisher-Neyman factorisation). □

Hence either feed the whitened residual $\mathrm{S}^{-1/2}\mathbf{r}$ (with the scene's whitening), or keep the learned model as the "score each family" box and leave scene-wise percentile calibration and the p^* decision after it, as the earlier proposal recommends. The second is safer: the calibration of Section 6 then holds whatever the network learned.

8.3 The generator-versus-reality gap

Let P be the real joint law. For any classifier h and a loss bounded in $[0, 1]$,

$$|R_P(h) - R_G(h)| \leq \mathrm{TV}(P, G).$$

The bound is loose but names the two parts of the gap:

- *Label shift.* π_P is dominated by "none" (about 50 false alarms per target among candidates). If only priors differ, $p_P(k \mid \boldsymbol{v}) \propto p_G(k \mid \boldsymbol{v})\,\pi_P(k)/\pi_G(k)$ [19]: correct the posterior, do not retrain.
- *Conditional shift.* Real false alarms (field edges, glint, shoreline, haze) are not generated by G, and real deployed materials are not library spectra (the domain gap of Section 7). This part cannot be corrected by reweighting. The cheapest partial fix is to take "none" examples from the real top-percentile candidates of training scenes, whose label noise (about 2% target) is tolerable, rather than from untouched pixels.

A classifier two-sample test measures the gap directly: train a discriminator to tell synthetic implants from real candidate pixels; an AUC well above 0.5 on held-out scenes means the namer will be trained on a world it will not see. A namer trained on linear mixtures and scored on linear-mixture plantings flatters itself by construction, which is why the earlier proposal's pass rule weights the Gudalur and oil sets.

8.4 As built (31 October 2026)

A namer was built on 31 October 2026 (`app/spectral_match/namer/`; design `docs/proposals/spectral-namer.md`, two levels: man-made or natural, then one of seven material classes). It departs from the plan above in four ways that bear on the propositions.

The generator. $a \sim$ log-uniform on $[0.05, 1]$ (not $[0.05, 0.5]$), so $P(a \le q) = \ln(q/0.05)/\ln 20$ and $\mathrm{E}[a] = 0.95/\ln 20 = 0.317$[C]; $\mathbf{t}$ a man-made training-split library spectrum (probability $\frac{1}{2}$), a natural one ($\frac{1}{4}$) or none ($\frac{1}{4}$, $a = 0$); a single-pixel linear mix with no PSF; $\mathbf{x} = \mathbf{G}(a\mathbf{t} + (1-a)\mathbf{b}) + \mathbf{n}$ with $\mathbf{G}$ diagonal, a clipped quadratic in the band index within ±10%, and $\mathbf{n}$ Gaussian with a per-tile, per-band σ equal to the RMS of horizontal neighbour differences over $\sqrt{2}$. The context $\mathbf{x}_C$ is summarised by the ring mean $\mathbf{r}$ over Chebyshev distances 3 to 7 (200 pixels), with $\mathbf{r}' = \mathbf{G}\mathbf{r}$, not by Heavy's $\hat{\mathbf{b}}$. Leakage (Section 5) therefore does not enter the namer; the conditional shift of the gap bound does, in full.

The input. $\boldsymbol{v} = (\mathbf{x}, \mathbf{r}')$ through Indradhanu V2's wavelength-aware stem (Heavy's global per-wavelength normalisation, then its post-stem BatchNorm, frozen at V2's running statistics so that training and evaluation normalise pixels, rings and library spectra alike), as $[\mathbf{z}_x, \mathbf{z}_r, \mathbf{z}_x - \mathbf{z}_r]$. The library keeps one entry per name, so a repeated name cannot leave a near-copy of the planted spectrum in its class. No scene covariance is supplied, so by Proposition 8.2 the namer cannot form the whitened statistics and its outputs cannot be calibrated across scenes on their own. The recommendation after Proposition 8.2 stands: keep scene-wise percentile calibration after the network. The inference API returns raw probabilities only.

The objective. Binary cross-entropy for M on every sample; $0.5\,|\hat{a} - a|$ on every sample; and, on man-made samples, cross-entropy over *class* scores $S_k = \ln \sum_{j \in k} e^{z_j}$, $z_j = \cos(\mathbf{z}, \boldsymbol{\omega}_j)/\tau$, over the training-split library with the planted entry's z_j set to $-\infty$. Since $e^{S_k} / \sum_{k'} e^{S_{k'}} = \sum_{j \in k} q_j$ for the entry softmax q, this is the class marginal of a per-entry softmax, and Proposition 8.1 applies at the class level: the minimiser is $p_{G'}(k \mid \boldsymbol{v}, M)$, where G' is the generator *with the planted entry removed from the candidate set.* That is the leave-one-out posterior, the quantity that generalises to spectra outside the library; with the entry kept, the minimiser would reward recognising the sample.

The test. Level-1 recall on held-out spectra at ACE's own natural false-alarm rate from the separability check (fill 0.25: $f = 0.166$, 0.215, 0.187 on PRISMA, EnMAP, AVIRIS-NG), required to exceed ACE's by 0.10 (bars 0.872, 0.764, 0.860)[C]; level-2 accuracy likewise, once ACE is re-scored on the seven classes; an open-set run holding one class (paint/coatings) out. The level-1 head is trained on planted man-made library spectra, so it is a two-class discriminator under G, not the lack-of-fit test of Section 6's open-set list. Results were pending when this section was written (run N001, H100, from 31 October 2026 11:30 UTC). Derivations at teaching pace: the step-by-step volume, chapter "The namer as built".

9 Measuring it

9.1 An OSM-labelled benchmark

The parked plan's judge is a benchmark built from OpenStreetMap tags over held scene footprints (`landuse=greenhouse_horticulture`, `plant:source=solar`, `surface=asphalt/concrete`, `roof:material=*`, harbour and water features). Let $\mathcal{O}_k$ be the objects of family k, $n_k = |\mathcal{O}_k|$. At a fixed coverage (share of objects answered) report per family

$$\mathrm{Acc}_k = \frac{1}{n_k} \sum_{o \in \mathcal{O}_k} \mathbf{1}[\hat{k}(o) = k], \qquad \mathbf{C}_{jk} = \frac{1}{n_j} \#\{o \in \mathcal{O}_j \,:\, \hat{k}(o) = k\},$$

the confusion matrix including the "unconfirmed" and "natural" outcomes, and the false-name rate on ordinary pixels and on false alarms separately (Remark 6.3). Methods are compared at equal coverage, as the LLD's rules require. If a fraction ϵ_L of OSM material tags is wrong and wrong tags are answered at chance among K families, the observed accuracy is $(1 - \epsilon_L)\mathrm{Acc}_k + \epsilon_L/K$, so differences between methods shrink by the factor $1 - \epsilon_L$ and the required sample grows by $1/(1 - \epsilon_L)^2$.

9.2 Paired bootstrap over objects

`app/evaluation/significance.py` resamples each scene's objects with replacement, the same draw for both methods, recomputes both measures and takes the 2.5–97.5 percentiles of the difference; background pixels are not resampled. The bench-resolution study notes that scene-level uncertainty is not covered, so the interval is a lower bound on the real uncertainty (`research/experiments/bench-resolution/README.md`)[M]. For per-family accuracy the same scheme applies with objects of family k in place of anomaly objects and $\mathbf{1}[\hat{k} = k]$ in place of "caught".

9.3 Minimum detectable improvement

Proposition 9.1 (Paired proportions)**.** *Let two methods be scored on the same n objects with discordance rate π_d (share of objects on which exactly one is right). The difference in accuracy has standard error $\approx \sqrt{\pi_d/n}$, and the smallest difference detectable with power $1-\beta$ at two-sided level α is*

$$\Delta_{\min} \approx (z_{1-\alpha/2} + z_{1-\beta})\sqrt{\pi_d/n}, \qquad n \approx (z_{1-\alpha/2} + z_{1-\beta})^2\, \pi_d/\Delta^2.$$

Proof. The per-object difference takes values ± 1 on discordant objects and 0 otherwise; its variance is $\pi_d - \Delta^2 \approx \pi_d$ for small Δ. Apply the normal approximation (McNemar's test). □

Worked example 9.2 (Sizing a family benchmark)**.** Check against the bench first: FTH1 vs FTB1 caught-top-1% intervals are about ± 1.9 points on 909 objects (`FTH1-heavy-joint/README.md`)[M], which by Proposition 9.1 means $\pi_d \approx 0.085$[C]. Two namers disagree more than two anomaly models; take $\pi_d = 0.2$[A]. With $z_{0.975} + z_{0.8} = 2.80$[C]:

objects per family n_k	30	100	300
$\Delta_{\min}$ at $\pi_d = 0.2$	23 pts	12.5 pts	7.2 pts
95% half-width of one accuracy near 0.5	18 pts	9.8 pts	5.7 pts

To resolve a 10-point per-family gain needs about 160 objects per family at $\pi_d = 0.2$ (80 at $\pi_d = 0.1$, 235 at 0.3)[C]. Objects from the same scene are correlated; with m objects per scene and intra-scene correlation ρ the effective size is $n/(1 + (m-1)\rho)$, which at $m = 20$, $\rho = 0.1$[A] is $n/2.9$. "Hundreds to thousands of objects per family", as the parked plan hopes for, is the right order; tens per family can only show large effects.

10 Open questions and the cheapest experiments

Ordered by cost; none needs a GPU instance beyond the laptop, and most need no training.

1. **Does RM's truncation discard what identifies families?** (Remark 2.4.) Implant library differences into bench tiles and measure the share of their whitened energy outside RM's top-30 residual subspace. CPU, an hour.
2. **How much does Heavy leak, and where?** (Section 5.) Implant library spectra at controlled fill, size and position relative to the token grid; run Heavy with and without the implant; estimate $\mathbf{L}_p$ from $\hat{\mathbf{b}}^{\text{with}} - \hat{\mathbf{b}}^{\text{without}}$. Gives l by size and position and the direction of leakage. Laptop GPU, a few hours.
3. **Background error against cell size.** Measure $\sigma_{e,4}$ and $\sigma_{e,8}$ along implanted target directions on FT04 and FT08 to test Proposition 5.6. Laptop GPU, an hour; it explains FT08's loss or refutes the explanation.

4. **Heavy's output floor.** Rerun `decoder-capacity/spectral_floor.py` on FTH3 epoch 5 ($D = 64$). Minutes.

5. **Residual-domain family matching.** In `match_families`, replace the ring mean by $\hat{\mathbf{b}}$ and the scene covariance by $\boldsymbol{\Sigma}_r$ (optionally projecting the floor and the learned-leakage directions out). Score on the planted set, Gudalur and oil under the LLD's protocol. CPU, about a day.

6. **Gain sensitivity.** Apply smooth ±5% and ±10% gains to the planted set and measure the drop in right names. Proposition 7.1 predicts it is comparable to family separation. CPU, hours.

7. **Lack-of-fit abstention and a candidate-conditional null.** Add test 3 of the open-set list and calibrate against the candidate population (Remark 6.3); measure false names on false alarms on the four PRISMA scenes. CPU, hours.

8. **OSM density.** The parked plan's first check: count OSM material-tagged objects inside held scene footprints by family, and compare with Worked example 9.2's 100–300 per family. Laptop, a few hours. It decides whether the benchmark can be the judge.

9. **Generator gap.** The classifier two-sample test of Section 8, synthetic implants against real candidates on held-out scenes. Laptop GPU, hours. Worth running before any namer is trained.

What would change the plan: if (2) shows $l \gtrsim 0.5$ for small objects, naming should use the ring background or a heavily masked pass, not the standard two-pass $\hat{\mathbf{b}}$; if (6) confirms a gain sensitivity at the level of family separation, the atmospheric-correction harmonisation is a prerequisite for naming, not a parallel track; if (8) finds tens rather than hundreds of objects per family, the real benchmark can only rank families coarsely and Gudalur and oil remain the gate.

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